

Novelty update changes explained

Let us assume the agent is at time $t = 1$, having just committed action a_1 , and is now evaluating a two-step candidate policy sequence: $\pi = (\hat{a}_2, \hat{a}_3)$.

State transitions

During policy evaluation, the agent predicts future states sequentially. Based on the Markov properties of the generative model, the expected future states are calculated as:

1. **Depth 1 State:** $q(\bar{s}_2 \mid a_1) = \mathbb{E}_{q(s_1)}[P(\bar{s}_2 \mid s_1, a_1)]$

2. **Depth 2 State:** $q(\bar{s}_3 \mid \hat{a}_2) = \mathbb{E}_{q(\bar{s}_2)}[P(\bar{s}_3 \mid \bar{s}_2, \hat{a}_2)]$

Notice that the transition into $\bar{s}_3$ is strictly parameterised by the candidate action $\hat{a}_2$.

Novelty calculation

When transition parameters are learned (Dirichlet updating), evaluating a policy yields an epistemic "novelty" reward. This represents the expected information gain about the transition matrix B . If we define the prior Dirichlet concentration parameters for a given action a as $b^{(a)}$, the expected state transition generates a pseudo-count update matrix Δb :

$$\Delta b = q(\bar{s}_{\tau+1}) \otimes q(\bar{s}_\tau)$$

The posterior parameters $b'^{(a)}$ are formed by adding this expected experience to the prior, scaled by a learning rate α :

$$b'^{(a)} = b^{(a)} + \alpha \Delta b$$

The novelty of the transition is then the Kullback-Leibler (KL) divergence between the updated posterior and the prior:

$$\mathcal{W} = D_{KL} [\text{Dir}(b'^{(a)}) \parallel \text{Dir}(b^{(a)})]$$

Issue in original implementation

The structural misalignment in the original code occurred because the algorithm decoupled the state transition from the Dirichlet parameter update. When evaluating the node for candidate action $\hat{a}_3$ at depth 2, the algorithm calculated the pseudo-count Δb using the transition that arrived at that node:

$$\Delta b = q(\bar{s}_3 \mid \hat{a}_2) \otimes q(\bar{s}_2 \mid a_1)$$

However, it then applied this pseudo-count to update the parameters of the *current* candidate action being evaluated ($\hat{a}_3$), rather than the action that caused the transition ($\hat{a}_2$). The flawed posterior was calculated as:

$$b'_{\text{flawed}}^{(\hat{a}_3)} = b^{(\hat{a}_3)} + \alpha [q(\bar{s}_3 \mid \hat{a}_2) \otimes q(\bar{s}_2 \mid a_1)]$$

This leads to the artificial divergence. If $\hat{a}_2$ is 'Defect' and $\hat{a}_3$ is 'Cooperate', the equation above states: "Update my beliefs about what happens when I Cooperate using the predicted consequences of Defecting". Because the predicted state of Defecting violates the prior expectations of Cooperating, the resulting D_{KL} is massive, heavily skewing the Expected Free Energy toward this contradictory sequence.

Corrected implementation

To preserve the causal structure, the novelty calculation must assign the expected information gain to the specific action that generated it. The corrected logic tracks the causal action and applies the pseudo-count Δb strictly to the parameters of $\hat{a}_2$:

$$b'_{\text{corrected}}^{(\hat{a}_2)} = b^{(\hat{a}_2)} + \alpha [q(\bar{s}_3 \mid \hat{a}_2) \otimes q(\bar{s}_2 \mid a_1)]$$

The novelty is then correctly evaluated as the information gained about the causal action:

$$\mathcal{W} = D_{KL} \left[\text{Dir}(b'_{\text{corrected}}^{(\hat{a}_2)}) \parallel \text{Dir}(b^{(\hat{a}_2)}) \right]$$

By ensuring that $b^{(a)}$ is only ever updated by Δb matrices generated under action a , the KL divergence accurately reflects genuine epistemic value, preventing the accumulation of artificial novelty and allowing the agent to adapt to changes in the payoff matrix.